

Lectures Review: Linear Models for Regression

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Outline

① What Regression Actually Estimates

② Least Squares and Normal Equations

③ Over-fitting, Priors, and Lineage

④ Summary

The formal question: which function should regression target?

$$\underbrace{\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^N \sim p(x, y)}_{\text{data from an unknown joint law}}, \quad R(g) := \mathbb{E}_{p(x, y)} [(y - g(x))^2] \quad g^* = \arg \min_g R(g)$$

- g : **any** candidate function we could pick (no linearity, no form assumed yet).
- g^* : the best possible predictor under squared loss — *we derive who it is, we do not assume it.*

Lemma (step 1): add and subtract the truth

Insert $f^*(x)$ (any f^* , to be chosen) via $y - g = (y - f^*) + (f^* - g)$ and expand the square:

$$R(g) = \mathbb{E}[(y - g(x))^2] = \underbrace{\mathbb{E}[(y - f^*(x))^2]}_{\text{noise: depends on } y, f^* \text{ only}} + \underbrace{2 \mathbb{E}[(y - f^*(x))(f^*(x) - g(x))]}_{\text{cross term} \quad \leftarrow \text{must be killed}} + \underbrace{\mathbb{E}[(f^*(x) - g(x))^2]}_{\text{distance from } g \text{ to } f^*, \geq 0}.$$

- Pure algebra: nothing assumed about f^* yet — the decomposition holds for *any* choice.
- Three roles: an irreducible noise term, a **cross term**, and a nonnegative distance from g to f^* .

Everything now hinges on killing the **cross term** — and *choosing* f^* is the only lever we have.

Lemma (step 2): the cross term vanishes iff $f^*(x) = \mathbb{E}[y | x]$

If choose $f^*(x) := \mathbb{E}[y | x]$. Then for *any* function $h(x)$:

$$\mathbb{E}[(y - f^*(x)) h(x)] = \mathbb{E}\left[\underbrace{\mathbb{E}[y - f^*(x) | x]}_{=0 \text{ at every } x} h(x)\right] = 0.$$

(**Tower property**: the residual $y - \mathbb{E}[y | x]$ has conditional mean zero at *every* x .)

If the **cross term** vanishes for *every* $h(x)$, take $h = \mathbb{E}[y | x] - f^*$; then $\mathbb{E}[(\mathbb{E}[y | x] - f^*)^2] = 0$, forcing $f^*(x) = \mathbb{E}[y | x]$ a.e.

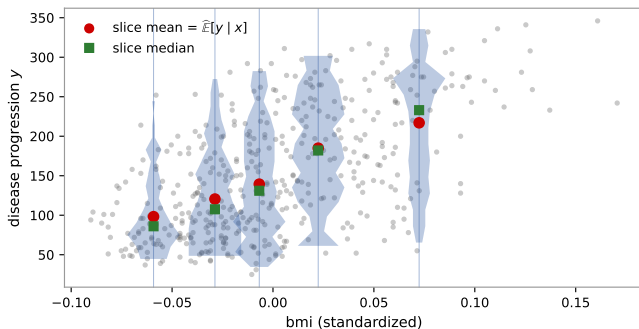
Conclusion: $g^*(x) = \mathbb{E}[y | x]$, and “noise” is what you failed to measure

$$R(g) = \underbrace{\mathbb{E}[(y - \mathbb{E}[y | x])^2]}_{\text{irreducible noise} = \mathbb{E}[\text{Var}(y|x)]} + \underbrace{2\mathbb{E}[(y - \mathbb{E}[y | x])(\mathbb{E}[y | x] - g(x))]}_{=0 \quad (\text{orthogonality, step 2})} + \underbrace{\mathbb{E}[(\mathbb{E}[y | x] - g(x))^2]}_{\geq 0, =0 \text{ iff } g = \mathbb{E}[y|x]}.$$

- The irreducible part is $\mathbb{E}[\text{Var}(y | x)]$: variance of y given *what you measured*.
- Noise is therefore not a property of the world — it is the part of y orthogonal to your chosen inputs.

MSE regression targets the conditional mean, full stop;
The noise floor of a problem is defined by the features, not by nature.

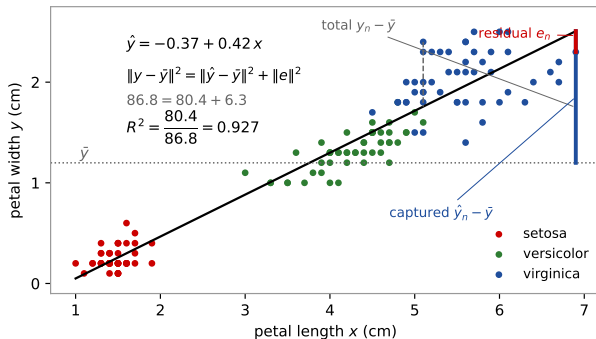
$\mathbb{E}[y | x]$: slice the scatter and watch the means



- Split bmi into vertical slices; each slice has its own little distribution (blue bands).
- Red dots: slice means — the empirical $\mathbb{E}[y | x]$; green: medians.

The regression curve is the locus of slice centers; “conditional expectation” is an operation you can watch, not a definition to memorize.

Living case: the Iris fit and what R^2 means



- At every point the vertical deviation splits: $y_n - \bar{y} = (\hat{y}_n - \bar{y}) + e_n$ — one segment, two roles.
- Squared and summed over N , the segments become vector lengths in $\mathbb{R}^{150}$ with $\mathbf{e} \perp \text{span}(\mathbf{X})$: Pythagoras gives the ledger $86.8 = 80.4 + 6.3$.
- $R^2 = \frac{80.4}{86.8} = 0.927$: **fraction of variance the projection captures.**

R^2 is the Pythagorean share of $\mathbf{y}$ that lies in $\text{span}(\mathbf{X})$.

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Recap: Geometric interpretation of Normal Equations

- ❶ **The problem now.** Restrict g to the linear family $g(x) = \mathbf{x}^\top \mathbf{w}$ and the population risk becomes a finite, concrete optimization:

$$\mathbf{w}^\star = \arg \min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 \quad \text{— again a projection, now in } \mathbb{R}^N.$$

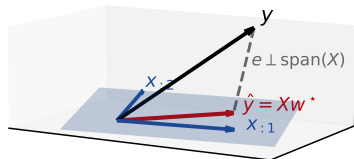
The normal equations are orthogonality, restated

$$\nabla \mathcal{L}(\mathbf{w}^\star) = \mathbf{0} \iff \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{w}^\star) = \mathbf{0} \iff \mathbf{e} \perp \text{span}(\mathbf{X})$$

- The matrix equation $\mathbf{X}^\top \mathbf{e} = \mathbf{0}$ is one row per feature: $\langle \mathbf{x}_{:d}, \mathbf{e} \rangle = 0$ for every $d = 1, \dots, D$.
- Consequence: the residual carries no linear signal that any feature could still explain.
- (FYI) This is the Orthogonal Projection Theorem.

Least squares = the L_2 projection of $\mathbf{y}$ onto the column space;
the gradient condition is orthogonality in coordinates.

The projection, drawn in the space where y lives



- Columns $x_{:1}, x_{:2}$ span a plane in $\mathbb{R}^N$; y floats above it.
- $\hat{y} = \mathbf{X}\mathbf{w}^*$ is the foot of the perpendicular; the dashed residual meets the plane at a right angle — *that is* $\mathbf{X}^T \mathbf{e} = \mathbf{0}$.

Fitting is dropping a perpendicular: the data span a floor, y floats above, the best linear answer is the shadow — its out-of-plane part is unrecoverable.

Existence and uniqueness of $\mathbf{w}$: a rank lemma with a one-line proof

$$\mathbf{X}^\top \mathbf{X} \mathbf{w} = \mathbf{X}^\top \mathbf{y} \text{ solvable uniquely} \iff \text{rank}(\mathbf{X}) = D$$

- Proof: $\mathbf{X}^\top \mathbf{X} \mathbf{v} = \mathbf{0} \Rightarrow \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 = 0 \Rightarrow \mathbf{X} \mathbf{v} = \mathbf{0}$; full column rank forbids nonzero $\mathbf{v}$.
- Failure modes: $D > N$ (more features than data) or (nearly) collinear columns.
- Consequence of non-uniqueness: infinitely many $\mathbf{w}$ give the same $\hat{\mathbf{y}}$ — the fit exists, the *coefficients* do not.

Invertibility of the Gram matrix is the question “can any two different coefficient vectors produce identical predictions?” — answered by the rank of $\mathbf{X}$.

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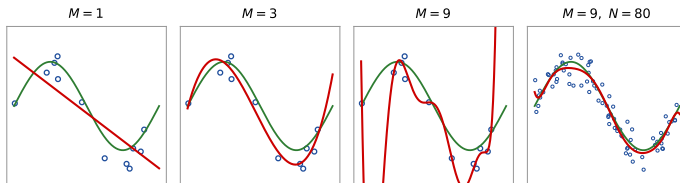
Feature augmentation: richer family, same linearity in $\mathbf{w}$

$$y_n \approx w_0 + w_1 x_n + \cdots + w_M x_n^M = \boldsymbol{\phi}(x_n)^\top \mathbf{w}, \quad \boldsymbol{\phi}(x) = [1, x, x^2, \dots, x^M]$$

- Nonlinear in x , **linear in $\mathbf{w}$** : every tool of the last three sections (projection, spectrum, ridge) survives unchanged — only the design matrix changes to $\Phi_{nj} = x_n^j$.
- “Linear model” refers to the parameters, not the curve.

Polynomial regression is linear regression after a change of features — the machinery is untouched, which is exactly why the failure it produces is instructive.

Living case: the polynomial sweep, read through df



- $M=1$: under-fit — approximation error dominates (the family cannot bend).
- $M=9, N=12$: $df \approx N$ — the fit copies the noise, direction by direction.
- $M=9, N=80$: same family, data regains authority — estimation error shrinks.

Under-fit and over-fit are the two summands of the excess risk; M moves one, N moves the other, and df keeps the score.

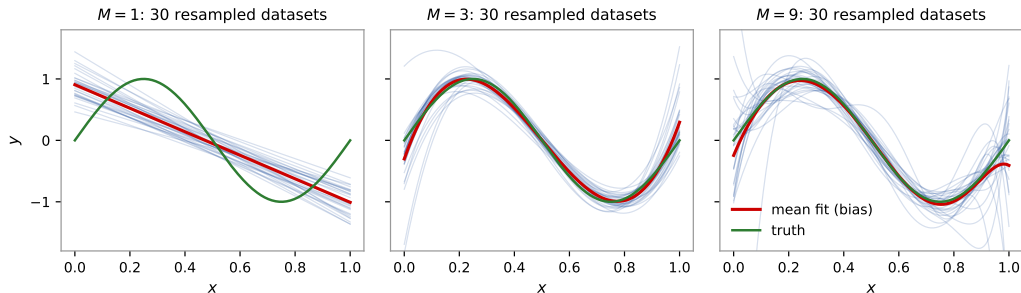
Bias–variance: the estimation error splits again

$$\mathbb{E}[(\hat{f}(x) - y)^2] = \underbrace{(\mathbb{E}\hat{f} - f^*)^2}_{\text{bias}^2} + \underbrace{\mathbb{E}(\hat{f} - \mathbb{E}\hat{f})^2}_{\text{variance}} + \underbrace{\text{Var}(y \mid x)}_{\text{irreducible}}$$

- Same Pythagoras move as Section 1, now over *resampled datasets*: $\hat{f}$ is random through $\mathcal{D}$.
- High- M fits: bias ≈ 0 , variance explodes — the wiggle between the points is dataset-to-dataset disagreement.

Over-fitting is variance: the model's sensitivity to which N points you happened to draw.
Regularization is buying bias to sell variance.

Bias and variance, drawn from 30 resampled datasets



$M=1$: the band is tight but off-target — bias. $M=9$: the mean (red) tracks the truth while individual fits whip around — variance. Over-fitting is the spread of the blue family, not any single bad curve.

Ridge pays its bias exactly where variance was loudest I

Expand $\mathbf{w} = \sum_i \alpha_i \mathbf{v}_i$ in the eigenbasis and substitute into the ridge normal equations $(\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I}) \mathbf{w} = \mathbf{X}^\top \mathbf{y}$:

$$(\sigma_i^2 + \lambda') \alpha_i = \mathbf{v}_i^\top \mathbf{X}^\top \mathbf{y} = \sigma_i \mathbf{u}_i^\top \mathbf{y} \quad \Rightarrow \quad \alpha_i^{\text{ridge}} = \underbrace{\frac{\sigma_i^2}{\sigma_i^2 + \lambda'}}_{f_i} \underbrace{\frac{\mathbf{u}_i^\top \mathbf{y}}{\sigma_i}}_{\alpha_i^{\text{OLS}}}$$

Under $\mathbf{y} = \mathbf{X} \mathbf{w}_{\text{true}} + \boldsymbol{\epsilon}$ the OLS coefficient is unbiased, $\mathbb{E}[\alpha_i^{\text{OLS}}] = \alpha_i^{\text{true}}$, so

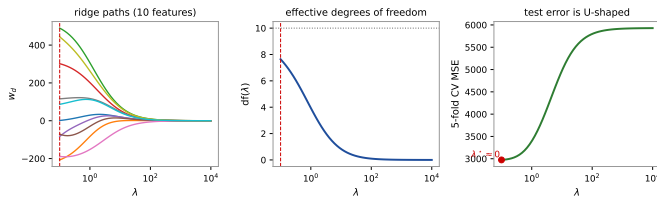
$$\text{bias}_i = \mathbb{E}[\alpha_i^{\text{ridge}}] - \alpha_i^{\text{true}} = (f_i - 1) \alpha_i^{\text{true}} = -\frac{\lambda'}{\sigma_i^2 + \lambda'} \alpha_i^{\text{true}}, \quad \text{Var}_i = f_i^2 \frac{\sigma^2}{\sigma_i^2}$$

Ridge pays its bias exactly where variance was loudest II

- Small- σ directions — where the OLS variance σ^2/σ_i^2 exploded — pay the most bias ($f_i \rightarrow 0$); strong directions are barely touched ($f_i \rightarrow 1$).
- The trade is *matched* to the pathology: variance removed exactly where it was created.

Ridge is not “smaller weights”; it is a direction-wise exchange of variance for bias.

Living case: cross-validation picks λ where the curve turns



- 5-fold CV MSE on Diabetes is U-shaped: left = too much bias, right = too much variance.
- The same λ^* (dashed) sits where the coefficient paths have shed the collinear serum block.
- Model selection = measuring the bias–variance trade, not arguing about it.

λ is chosen by held-out error — the only honest witness, since training error votes for $\lambda = 0$ by construction.

MAP: where does the penalty come from? I

The question. Section 3 introduced $\lambda \|\mathbf{w}\|^2$ to repair the inversion — a fix that works, but came from algebra, not from a principle. Where *should* a penalty come from, and what does λ mean?

The Bayesian move. Treat $\mathbf{w}$ as uncertain *before* seeing data, and encode that belief as a distribution $p(\mathbf{w})$. Bayes's rule combines belief with evidence:

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto \underbrace{p(\mathbf{y} \mid \mathbf{X}, \mathbf{w})}_{\text{likelihood: what the data say}} \underbrace{p(\mathbf{w})}_{\text{prior: what we believed}}$$

The *maximum a posteriori* (MAP) estimate is the $\mathbf{w}$ that maximizes the right-hand side — the MLE, corrected by belief.

MAP: where does the penalty come from? II

One factoring step. Take Gaussian noise (Section 2) and a Gaussian belief $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I})$ — “each weight is probably near zero, with scale τ ”. Taking $-\log$ turns the product into a sum:

$$-\log p(\mathbf{y} \mid \mathbf{X}, \mathbf{w}) = \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \text{const}, \quad -\log p(\mathbf{w}) = \frac{1}{2\tau^2} \|\mathbf{w}\|^2 + \text{const}$$

$$\mathbf{w}_{\text{MAP}} = \arg \min_{\mathbf{w}} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 + \frac{1}{2\tau^2} \|\mathbf{w}\|^2 \quad \Longleftrightarrow \quad \text{ridge with } \lambda' = \frac{\sigma^2}{\tau^2}$$

Reading the identification.

- The penalty strength is a *noise-to-belief ratio*: noisy data (σ^2 large) $\Rightarrow$ listen to the prior; vague prior (τ^2 large) $\Rightarrow$ listen to the data.

MAP: where does the penalty come from? III

- $\tau^2 \rightarrow \infty$ (flat prior) recovers MLE/OLS; small τ asserts “weights are small” so strongly that only overwhelming data can override it.
- Nothing was added by hand: the $\|\mathbf{w}\|^2$ term *is* the Gaussian prior, after $-\log$.

Ridge is Bayes’s rule with a Gaussian belief about weights — the regularizer is an epistemic statement, not a numerical trick.

The posterior buys what no penalty can: uncertainty I

The question. MAP and ridge deliver the same *point* $\mathbf{w}_{\text{ridge}}$. But a point is silent about its own error — how wrong could it be? No choice of penalty can answer that, because a penalty is only an objective to minimize. The Bayesian reading can, because it keeps the whole distribution $p(\mathbf{w} \mid \mathbf{X}, \mathbf{y})$, not just its maximizer.

The computation: a product of Gaussians is Gaussian. Both factors are exponentials of quadratics in $\mathbf{w}$:

$$p(\mathbf{w} \mid \mathbf{X}, \mathbf{y}) \propto \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|^2 - \frac{1}{2\tau^2} \|\mathbf{w}\|^2\right) = \exp\left(-\frac{1}{2\sigma^2} [\mathbf{w}^\top (\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I}) \mathbf{w} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y}] + \text{const}\right)$$

Completing the square in $\mathbf{w}$ reads off a Gaussian, with center and curvature visible in the bracket:

$$\mathbf{w} \mid \text{data} \sim \mathcal{N}\left(\underbrace{(\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}}_{= \mathbf{w}_{\text{ridge}}}, \underbrace{\sigma^2 (\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I})^{-1}}_{\text{curvature}^{-1}}\right)$$

The posterior buys what no penalty can: uncertainty II

The posterior mean is exactly the ridge solution — MAP and posterior mean coincide because a Gaussian is symmetric.

Reading the covariance, direction by direction. In the eigenbasis the covariance is diagonal:

$$\text{posterior std along } v_i = \frac{\sigma}{\sqrt{\sigma_i^2 + \lambda'}}$$

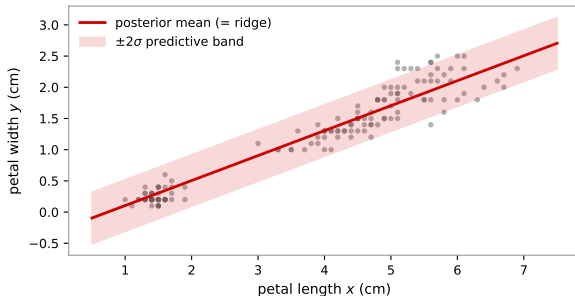
- Strong directions (σ_i large): the posterior is tight — data pinned the coefficient down.
- Weak directions (σ_i small): the posterior stays wide, floored only by the prior at $\sigma/\sqrt{\lambda'} = \tau$ — exactly the directions where Section 3 found the OLS variance σ^2/σ_i^2 exploding. The spectrum prices uncertainty the same way it priced error.

The posterior buys what no penalty can: uncertainty III

- Prediction inherits the uncertainty: $\hat{y}_{\text{new}} = \mathbf{x}^T \mathbf{w}$ is a linear function of a Gaussian, so it comes with an interval, not just a point — drawn on the next slide.

MAP and ridge agree on the point; only the Bayesian reading tells you how wrong the point could be.

What a posterior looks like in practice: error bars on Iris



- Bayesian ridge on petal length $\rightarrow$ width: the posterior mean is the ridge line.
- The band $= \sigma \sqrt{1 + x^\top (\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I})^{-1} x}$: narrowest at the data's center, flaring at the edges.
- The flare *is* the covariance $\sigma^2 (\mathbf{X}^\top \mathbf{X} + \lambda' \mathbf{I})^{-1}$ made visible.

A prediction without a band is half an answer: the posterior tells you where the data ran out — off-center, the honest answer widens.

The lineage: one object, two souls

- **Geometry** (Legendre 1805): least squares as the projection minimizing residual length — algebra, no probability.
- **Inference** (Gauss 1809): the same solution as the most probable explanation under normal errors; then Gauss–Markov (BLUE), Fisher (MLE, information), Cramér–Rao.
- **Robustness and posedness** (Huber 1964; Tikhonov): change the noise geometry, or constrain the inverse — MAE/Huber and ridge are the same move in two vocabularies.
- **Principles above algorithms** (Jaynes): maximum entropy says the noise laws themselves are forced by what you claim to know.

Linear regression is where projection (linear algebra), conditional expectation (probability), KL projection (statistics), and spectral dynamics (optimization) are provably one object — learn it here once, recognize it everywhere.

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Five checks, one per root

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- ❸ **What does the spectrum say?** (linear algebra) Read $\sigma_i(\mathbf{X})$ and κ before trusting $\mathbf{w}^\star$; $\text{df} = \sum_i \sigma_i^2 / (\sigma_i^2 + \lambda)$ is the real complexity.

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- 5 **Point estimate or belief?** (inference) MAP gives a point; the posterior gives error bars — know which one you need.

One object, four lenses: projection = conditional expectation = KL projection = spectral contraction — later lectures re-run this chain with a new family or a new loss.